# MICROSOFT EXCEL: THE COMPLETE PRACTICAL LEARNING WORKBOOK

INTENSIVE PRACTICE EDITION

Beginner to Advanced | Real-World Scenarios | Hands-On Practice

Designed for Students, MIS Executives, Finance Professionals, and Data Analysts in India. Includes expanded practice workshops, real datasets, advanced formulas, Power Query, Macros, and full Interview Preparation.

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# Part 1: Excel Interface & Cell Referencing

**What it is:** Excel is a grid of columns (letters A, B, C) and rows (numbers 1, 2, 3). The intersection is a Cell (e.g., A1). The true power of Excel lies in its ability to copy formulas across thousands of rows automatically. To do this without errors, you must master **Cell Referencing**.

## The Golden Rule of Referencing ($)

When you drag a formula down, Excel automatically shifts the row numbers (A1 becomes A2). When you drag it right, it shifts the columns (A1 becomes B1). The `$` symbol **locks** the row or column. You can cycle through locks by pressing the **F4** key while typing a cell address.

| Type | Syntax | What happens when dragged down? | What happens when dragged right? |
|---|---|---|---|
| Relative | `A1` | Changes to A2, A3, A4... | Changes to B1, C1, D1... |
| Absolute | `$A$1` | Stays $A$1 (Locked) | Stays $A$1 (Locked) |
| Mixed (Row Lock) | `A$1` | Stays A$1 (Row 1 is locked) | Changes to B$1, C$1... |
| Mixed (Col Lock) | `$A1` | Changes to $A2, $A3... | Stays $A1 (Col A is locked) |

**Practice Workshop: Referencing**

**Scenario:** You are calculating employee bonuses. The base salary is in Column B. The company-wide bonus percentage (15%) is stored in cell **E1**.

| Row | A (Emp Name) | B (Salary ₹) | C (Bonus ₹) | D (Total ₹) | E (Settings) |
|---|---|---|---|---|---|
| 1 | *Headers* | *Headers* | *Headers* | *Headers* | **15%** |
| 2 | Rahul | 50,000 | *formula here* | *formula here* | |
| 3 | Priya | 65,000 | *formula here* | *formula here* | |
| 4 | Amit | 45,000 | *formula here* | *formula here* | |

**Task 1:** Write the formula for Rahul's Bonus in C2, so that it can be dragged down to C3 and C4 without breaking.
**Incorrect Formula:** `=B2*E1`. (If dragged down to C3, it becomes `=B3*E2`. E2 is empty, so Priya gets ₹0 bonus!).
**Correct Formula:** `=B2*$E$1`. (B2 changes to B3, but $E$1 stays locked on the 15%).
**Task 2:** Write the formula for Total in D2.
**Formula:** `=B2+C2`. (No locks needed, as both salary and bonus should move down row by row).

**Common Error: #VALUE!**

If you see #VALUE! when multiplying, one of your cells contains text instead of a number. Check if "50,000" was typed with a space, or if E1 says "15 percent" instead of the mathematical number `15%`.

## Part 2: Custom Formatting & Data Types

**Why it matters:** Excel stores the *value* and the *format* separately. You can make a cell display "₹ 50,000 Cr" while the underlying math just sees "50000". If you type the letters "Cr" manually into the cell, Excel treats it as text, and your SUM formulas will fail.

### Essential Custom Number Formats (Ctrl + 1)

Select your cells, press `Ctrl + 1`, go to the **Number** tab, and select **Custom**. Type the codes below:

| Requirement | Custom Code to Type | If you type `1250000`, it displays as: |
|---|---|---|
| Indian Comma System (Lakhs/Crores) | `##\,##\,##0` | 12,50,000 |
| Adding text without breaking math | `0 "Boxes"` | 1250000 Boxes |
| Hiding zeroes (Display blank instead of 0) | `0;-0;;@` | *(Leaves cell visually blank if value is 0)* |
| Format Phone Numbers (+91) | `+91-00000 00000` | +91-12500 00000 |

**Practice Workshop: Data Cleaning via Formatting**

**Scenario:** You download a bank statement. The dates look like "45210" and the amounts look like "5000.5". You need to make it presentable for the CFO.

**Task 1: Fix the Dates**

1. Select the Date column.
2. Press `Ctrl + Shift + #` (Quick shortcut for Date format).
3. *Alternative:* Press `Ctrl + 1`, select Custom, and type `dd-mmm-yyyy`. The number 45210 becomes **11-Oct-2023**.

**Task 2: Fix the Currency**

1. Select the Amount column.
2. Press `Ctrl + Shift + $`. It will format to your regional currency (e.g., ₹5,000.50).
3. To remove the decimals, click the "Decrease Decimal" button on the Home Ribbon twice. Result: **₹5,001** (Excel visually rounds it, but keeps 5000.5 in memory).

**Interview Relevance**

**Q:** "I have a list of Employee IDs that must be 5 digits (e.g., 00123). But when I type 00123, Excel changes it to 123. How do I fix this?"
**A:** "There are two ways. 1) Type an apostrophe before the number ( `'00123` ) to force Excel to treat it as text. 2) Select the column, press Ctrl+1, go to Custom format, and type `00000`. This forces a 5-digit display with leading zeroes while keeping it as a number."

## Part 3: Logical Functions (IF, AND, OR, IFS)

**What it is:** Logical functions allow Excel to make decisions based on your data. Instead of manually checking if every student passed, Excel does it instantly.

### 1. The Standard IF Function

**Syntax:** `=IF(logical_test, value_if_true, value_if_false)`

*Example:* `=IF(B2>=40, "Pass", "Fail")`

### 2. Combining Logic (AND / OR)

Sometimes you need multiple conditions to be true. You nest `AND()` or `OR()` inside the IF.

- **AND:** All conditions must be met. `=IF(AND(B2>=40, C2="Yes"), "Pass", "Fail")` (Score >= 40 AND Attendance = Yes).
- **OR:** Only one condition needs to be met. `=IF(OR(B2="Manager", B2="Director"), "Approve", "Reject")`.

### 3. Multiple Conditions (Nested IF vs IFS)

If you have 3 or more outcomes (e.g., Grades A, B, C, F), you can use the modern `IFS()` function (Office 2019+).

**Syntax:** `=IFS(condition1, result1, condition2, result2, TRUE, catch_all_result)`

**Practice Workshop: Sales Commission Logic**

**Dataset (A1:C5):**

| Row | A (Sales Rep) | B (Revenue ₹) | C (New Client?) |
|---|---|---|---|
| 2 | Karan | 1,20,000 | Yes |
| 3 | Sita | 85,000 | No |
| 4 | Ravi | 45,000 | Yes |
| 5 | Pooja | 2,10,000 | No |

**Task 1: Basic Bonus**
If Revenue is over 1,00,000, they get a "Bonus", otherwise "No Bonus".
*Formula in D2:* `=IF(B2>100000, "Bonus", "No Bonus")`. (Karan & Pooja get Bonus).
**Task 2: The "AND" Condition**
A rep only gets a "Super Bonus" if Revenue is over 1,00,000 **AND** they brought a New Client ("Yes").
*Formula in E2:* `=IF(AND(B2>100000, C2="Yes"), "Super Bonus", "-")`. (Only Karan gets it).
**Task 3: Tiered Commission using IFS**
Assign commission rates: Revenue > 2,00,000 = 10%. Revenue > 1,00,000 = 5%. Below 1L = 2%.
*Formula in F2:* `=IFS(B2>200000, 10%, B2>100000, 5%, TRUE, 2%)`.
*Note:* IFS evaluates from left to right. Always put the highest/strictest condition first!

## Part 4: Conditional Math (SUMIFS, COUNTIFS)

**Why it matters:** MIS reporting is almost entirely based on aggregating data based on criteria. "What were total sales in Mumbai?" "How many laptops did we sell in Q1?" The `SUMIFS` and `COUNTIFS` functions are your primary tools.

### The Syntax

Always use the plural version (`SUMIFS` instead of `SUMIF`). It handles both single and multiple criteria effortlessly.

- `=SUMIFS(sum_range, criteria_range1, criteria1, [criteria_range2, criteria2]...)`
- `=COUNTIFS(criteria_range1, criteria1, [criteria_range2, criteria2]...)`

**Master Practice Workshop: Regional Sales Analysis**

**Raw Data Dataset (A1:E7):**

| Row | A (Date) | B (Region) | C (Product) | D (Units) | E (Revenue ₹) |
|---|---|---|---|---|---|
| 2 | 01-Oct | West | Laptop | 5 | 2,50,000 |
| 3 | 05-Oct | North | Mouse | 120 | 60,000 |
| 4 | 12-Oct | West | Monitor | 10 | 80,000 |
| 5 | 15-Oct | South | Laptop | 3 | 1,50,000 |
| 6 | 20-Oct | West | Laptop | 2 | 1,00,000 |
| 7 | 25-Oct | North | Monitor | 5 | 40,000 |

**Challenge 1: Single Condition Sum**
Calculate total Revenue generated in the "West" region.
*Logic:* Sum Column E, where Column B is "West".
*Formula:* `=SUMIFS(E2:E7, B2:B7, "West")`. **Result: ₹4,30,000**
**Challenge 2: Multiple Conditions Sum (AND logic)**
Calculate total Revenue for "Laptops" sold in the "West" region.
*Logic:* Sum Column E, where Col B is "West" AND Col C is "Laptop".
*Formula:* `=SUMIFS(E2:E7, B2:B7, "West", C2:C7, "Laptop")`. **Result: ₹3,50,000**
**Challenge 3: Using Mathematical Operators**
Count how many orders had a Revenue greater than ₹90,000.
*Logic:* Count Column E where value is > 90000. Operators must be in quotes.
*Formula:* `=COUNTIFS(E2:E7, ">90000")`. **Result: 3 orders**
**Challenge 4: Using Cell References for Criteria**
Instead of hardcoding "West", you type West in cell **G2**. Write a dynamic formula.
*Formula:* `=SUMIFS(E2:E7, B2:B7, G2)`. (Now you can change G2 to North, and the sum updates instantly).

## Part 5: Lookup Functions (VLOOKUP Deep Dive)

**What it is:** VLOOKUP (Vertical Lookup) searches for a unique ID in the first column of a table, and returns data from the same row in another column.

### The 4 Arguments of VLOOKUP

`=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

1. **lookup_value:** What are you looking for? (e.g., Emp ID "E105").
2. **table_array:** Where is the data table? (The lookup value MUST be in the far left column of this range).
3. **col_index_num:** Which column number has the answer? (Counting from 1, left to right, within your selected table).
4. **range_lookup:** Type `0` or `FALSE` for an Exact Match. Type `1` or `TRUE` for an Approximate Match. (99% of the time, you use 0).

**Practice Workshop: Exact Match VLOOKUP**

**Master Data Table (Located on Sheet2, Range A1:D5):**

| A (Prod Code) | B (Product Name) | C (Category) | D (Price ₹) |
|---|---|---|---|
| P001 | ThinkPad T14 | Laptop | 85,000 |
| P002 | Logitech M330 | Accessory | 1,200 |
| P003 | Dell U2720Q | Monitor | 45,000 |
| P004 | MacBook Air | Laptop | 92,000 |

**Your Task (On Sheet1):**
You have an invoice form. In cell **A2**, you type "P003". Write formulas in B2 and C2 to fetch the Product Name and Price.
**Formula for Product Name (B2):**
`=VLOOKUP(A2, Sheet2!$A$1:$D$5, 2, 0)`
*Explanation:* Find A2 ("P003") in the Master Data. Return column 2 (Product Name). Require exact match (0).
**Result: Dell U2720Q.**
**Formula for Price (C2):**
`=VLOOKUP(A2, Sheet2!$A$1:$D$5, 4, 0)`
*Explanation:* Same logic, but return column 4 (Price). **Result: 45,000.**

**Top 3 Reasons VLOOKUP Returns #N/A**

1. **No Locking:** You forgot to use `$` on your table_array, so when you dragged the formula down, the search box shifted down and missed the data. Always use `$A$1:$D$100`.
2. **Extra Spaces:** "P003 " (with a space) does not equal "P003". Fix with `=TRIM(A2)`.
3. **Wrong Direction:** VLOOKUP can only search from left to right. It cannot find a Product Name and return a Prod Code to its left.

# Part 6: Advanced Lookups (Approximate & Two-Way)

## 1. Approximate Match VLOOKUP (range_lookup = 1)

Used for sliding scales, like tax brackets, grading systems, or tiered commissions. **Crucial Rule:** The lookup column MUST be sorted in ascending order (smallest to largest).

**Practice Workshop: Grading System**

**Tax/Grade Table (G1:H5):**

| G (Score Min) | H (Grade) |
|---|---|
| 0 | Fail |
| 40 | Pass |
| 60 | First Class |
| 80 | Distinction |

**Task:** A student scores 65 in cell A2. Find their grade.
*Formula:* `=VLOOKUP(A2, $G$1:$H$5, 2, 1)`
*How it works:* It looks for 65. It hits 60, then sees the next tier is 80. Since 65 is less than 80, it drops back to the 60 tier and returns **"First Class"**.

## 2. INDEX and MATCH (The VLOOKUP Alternative)

Before XLOOKUP, this was the professional standard. It overcomes VLOOKUP's biggest flaw: it can look left.

- `MATCH(lookup_value, lookup_array, 0)`: Tells you the ROW NUMBER where data lives.
- `INDEX(return_array, row_num)`: Returns the value at that row number.

**Practice Workshop: Looking Left**

**Data (A1:C4):** A=Dept, B=EmpID, C=Name. (Notice EmpID is in the middle!).
**Task:** Look up EmpID "E105" (in cell F2) and return their Department (from Col A).
*Formula:* `=INDEX(A1:A4, MATCH(F2, B1:B4, 0))`
*Logic:* MATCH finds what row "E105" is in Col B. INDEX then extracts the value from that exact row in Col A.

## 3. XLOOKUP (Microsoft 365)

The modern replacement for VLOOKUP, HLOOKUP, and INDEX/MATCH. It is simpler, safer, and faster.

**Syntax:** `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])`

*Task:* Same as above (Find Dept based on EmpID).
*XLOOKUP Formula:* `=XLOOKUP(F2, B1:B4, A1:A4, "Not Found")`

## Part 7: Text Functions & Data Cleaning Workshop

**Why it matters:** Data exported from ERPs (SAP, Oracle) is often messy. You cannot run PivotTables or VLOOKUPs on dirty data. Data Analysts spend 70% of their time cleaning text.

### Essential Text Extractors

| Function | Syntax | Action | Result on "IND-9921-MH" |
|---|---|---|---|
| **LEFT** | `=LEFT(A2, 3)` | Takes first 3 chars from left. | IND |
| **RIGHT** | `=RIGHT(A2, 2)` | Takes last 2 chars from right. | MH |
| **MID** | `=MID(A2, 5, 4)` | Start at char 5, extract 4 chars. | 9921 |
| **LEN** | `=LEN(A2)` | Counts total characters. | 11 |

**Practice Workshop: Advanced Data Cleaning**

**Raw Data in Column A:**

- Row 2: `  rahul.sharma@company.com  ` (Has leading/trailing spaces)
- Row 3: `TXN_8472_FAILED`
- Row 4: `Mumbai, Maharashtra`

**Challenge 1: Clean Spaces and Capitalize Names**
You need to extract the email, remove spaces, and make it lowercase.
*Solution:* `=LOWER(TRIM(A2))` **Result: rahul.sharma@company.com**
**Challenge 2: Substitute Unwanted Characters**
Turn "TXN_8472_FAILED" into "TXN-8472-FAILED" (Replace underscores with hyphens).
*Solution:* `=SUBSTITUTE(A3, "_", "-")`
**Challenge 3: Dynamic Extraction (TEXTBEFORE / TEXTAFTER)**
For "Mumbai, Maharashtra", extract just the City (everything before the comma).
*Solution (O365):* `=TEXTBEFORE(A4, ",")` **Result: Mumbai**
*Solution (Legacy):* `=LEFT(A4, FIND(",", A4)-1)` (Finds position of comma, extracts up to that -1).

**Interview Question: "Numbers stored as Text"**

**Q:** You export a report and your SUM formula returns 0, even though you see numbers. They have a small green triangle in the corner. How do you fix this?
**A:** The numbers are stored as text. I can fix this by: 1) Highlighting the column, clicking the warning icon, and selecting 'Convert to Number'. 2) Using Data > Text to Columns and clicking Finish. 3) Multiplying the column by 1 using Paste Special, or wrapping the cell in `=VALUE(A2)`.

## Part 8: Date Math & Deadlines Workshop

**What it is:** Excel dates are not text; they are numbers representing days since Jan 1, 1900. Because they are numbers, you can do math on them (Date + 30 days = New Date).

### Essential Date Functions

- `=TODAY()`: Returns today's date. (Updates every time you open the file).
- `=EOMONTH(start_date, months)`: Returns the End Of the Month. E.g., `=EOMONTH("15-Jan-2023", 0)` gives 31-Jan-2023.
- `=EDATE(start_date, months)`: Adds exact months. E.g., `=EDATE("15-Jan-2023", 1)` gives 15-Feb-2023.
- `=NETWORKDAYS(start, end, [holidays])`: Counts only Mon-Fri working days between two dates.

> **Practice Workshop: HR & Finance Trackers**
>
> **Scenario 1: Invoice Aging (Finance)**
> An invoice was generated on **10-Oct-2023** (Cell A2). Payment terms are 45 days. Today is 01-Dec-2023.
> **Task A: Calculate Due Date.**
> *Formula:* `=A2 + 45`. **Result: 24-Nov-2023** (Format cell as Date).
> **Task B: Calculate Days Overdue from Today.**
> *Formula:* `=TODAY() - (A2 + 45)`. **Result: 7** (Format cell as General Number). If negative, it is not due yet.
> **Scenario 2: Employee Tenure (HR)**
> Ravi joined on **01-Mar-2020** (Cell B2). You need to calculate his exact years of service as of today.
> **Task C: Calculate Years of Service.**
> *Formula:* `=DATEDIF(B2, TODAY(), "Y")`
> *Note:* DATEDIF is a hidden function in Excel. You won't see tooltips for it, but it works perfectly. Use "Y" for Years, "M" for Months, "D" for Days.
> **Scenario 3: SLA / Deadline Calculation (Operations)**
> A customer logs a ticket on **05-Oct-2023** (Cell C2). Your team has 5 *working days* to resolve it.
> **Task D: Calculate Deadline Date.**
> *Formula:* `=WORKDAY(C2, 5)`. Excel automatically skips weekends (Sat/Sun) to give the true deadline date.

# Part 9: Excel Tables & Data Validation

## 1. Excel Tables (Ctrl + T)

Never work with raw grids of data. Click inside your data and press `Ctrl + T` to create an official Excel Table. This provides massive benefits:

- **Auto-Formatting:** Banded rows (alternate colors) apply automatically.
- **Auto-Expanding Ranges:** When you add a new row of data at the bottom, the Table expands. Any charts or PivotTables connected to this table will automatically include the new data upon refresh (No more updating data source ranges!).
- **Auto-Filling Formulas:** Type a formula in row 1, hit Enter, and it instantly copies to row 10,000.
- **Total Row:** Go to Table Design > Check "Total Row". A row appears at the bottom with dropdowns for SUM, AVERAGE, COUNT.

## 2. Data Validation (Error Prevention)

If your formula breaks because someone typed "Mmbai" instead of "Mumbai", you need Data Validation.

> **Practice Workshop: Building a Bulletproof Form**
>
> **Task 1: Create a Dropdown List**
>
> 1. Select the cells where users will enter the City (e.g., B2:B100).
> 2. Go to Data > Data Validation.
> 3. Under "Allow", select **List**.
> 4. In the "Source" box, type: `Mumbai, Delhi, Bengaluru, Chennai`. (Or select a range containing these names). Click OK.
>
> **Task 2: Restrict Data to Past Dates Only (e.g., Date of Birth)**
>
> 1. Select the DOB column (C2:C100).
> 2. Data > Data Validation.
> 3. Allow: **Date**.
> 4. Data: **Less than**.
> 5. End Date: `=TODAY()`. Click OK. (Users can no longer enter future dates).
>
> **Task 3: Add Custom Error Alerts**
>
> 1. In the Data Validation menu, click the **Error Alert** tab.
> 2. Title: "Invalid Date". Error Message: "You cannot be born in the future. Please check the year."

# Part 10: Conditional Formatting Workshop

**What it is:** Changing the color of a cell automatically based on its value. Used extensively in Dashboards and MIS Trackers to highlight exceptions (e.g., low stock, high attrition, overdue tasks).

## 1. Standard Highlighting Rules (Home > Conditional Formatting)

- **Highlight Duplicates:** Select EmpID Column > Highlight Cells Rules > Duplicate Values. (Crucial for finding data entry errors).
- **Top/Bottom Rules:** Select Sales Column > Top/Bottom Rules > Top 10%. (Highlights top performers green).
- **Data Bars:** Select Revenue Column > Data Bars > Blue. (Creates a mini bar-chart directly inside the cell based on the value).

**Practice Workshop: Advanced Formula-Based Formatting**

Standard rules format the single cell containing the number. *Formula-based formatting* allows you to format an **entire row** based on a status in one column.
**Scenario:** You have a project tracker (Columns A to E). Column D contains the "Status" (Pending, In Progress, Complete). You want the entire row to turn Green if Column D is "Complete".
**Step-by-Step Solution:**

1. **Crucial First Step:** Select your ENTIRE data table, excluding headers (e.g., `A2:E100`). Ensure the active cell is A2.
2. Go to Home > Conditional Formatting > New Rule.
3. Select **"Use a formula to determine which cells to format"**.
4. In the formula box, type: `=$D2="Complete"`.
   *Logic Check:* The `$` locks Column D, so when Excel checks cell A2, it looks at D2. When it checks cell B2, it still looks at D2. This forces the whole row to react to Col D. The row number (2) is NOT locked, so it moves down to check row 3, 4, etc.
5. Click the **Format** button, go to the Fill tab, select Green. Click OK twice.

**Troubleshooting Conditional Formatting**

If your entire table turns green randomly, or rows are mismatched by one, you made a referencing error. Go to Conditional Formatting > Manage Rules. Check the "Applies to" range (e.g., A2:E100). Check the formula. If "Applies to" starts at A2, your formula MUST reference row 2 (e.g., `$D2`). If they are misaligned, the highlighting will be broken.

# Part 11: PivotTables (The Analysis Engine)

**Why it matters:** A PivotTable takes 500,000 rows of raw transactional data and summarizes it into a clean, readable report in 3 seconds, with zero formulas required. It is the most important tool in Excel.

## The Golden Rules of Source Data

Before inserting a PivotTable, your data MUST follow these rules:

1. Every column must have a unique header name. No blank headers.
2. No totally blank columns or rows breaking the data block.
3. No subtotals mixed into the raw data.
4. Format the data as an Excel Table (`Ctrl + T`) before inserting the Pivot.

> **Practice Workshop: Building and Modifying PivotTables**
>
> **Scenario:** You insert a PivotTable based on 5 years of retail sales data (Date, Store, Category, Profit).
> **Task 1: Total Profit by Category**
> *Action:* Drag the `Category` field into the **Rows** box. Drag the `Profit` field into the **Values** box.
> **Task 2: Breaking it down by Store (Cross-Tabulation)**
> *Action:* Drag the `Store` field into the **Columns** box. You now have a matrix (Category on left, Stores across the top).
> **Task 3: Percentage of Total (No formulas needed!)**
> Your manager wants to see the % contribution, not the raw rupees.
> *Action:* Right-click any number inside the PivotTable. Go to **Show Values As** > **% of Grand Total**. The table instantly converts to percentages.
> **Task 4: Grouping Dates (Daily to Monthly/Yearly)**
> You dragged 'Date' into Rows, but it shows every single day (1-Jan, 2-Jan...).
> *Action:* Right-click any Date in the PivotTable. Select **Group**. Choose **Months** and **Years**. Click OK. Excel rolls up thousands of daily transactions into neat monthly summaries.

> **Slicers (Interactive Dashboards)**
>
> Instead of using the clunky Filter box, click inside your PivotTable, go to PivotTable Analyze ribbon, and click **Insert Slicer**. Select 'Region'. A visual button panel appears on your sheet. Clicking "North" filters the PivotTable instantly. This is the foundation of interactive dashboards.

# Part 12: Advanced PivotTables & Charts

## Calculated Fields in PivotTables

Sometimes your raw data has *Revenue* and *Cost*, but missing *Profit*. You should NOT add a formula next to the PivotTable (it will break when the table updates). Instead, build the formula *inside* the PivotTable.

**Practice: Adding a Calculated Field**

1. Click inside your PivotTable.
2. Go to PivotTable Analyze > Fields, Items, & Sets > **Calculated Field**.
3. Name: `Profit Margin`
4. Formula box: Clear the zero, then double-click your fields to write: `=(Revenue - Cost) / Revenue`.
5. Click Add, then OK. The new field appears in your PivotTable perfectly aggregated.

## Data Visualization: Best Practices

To add a chart, click the PivotTable, go to Insert > **PivotChart**. Which chart should you choose?

| Business Question | Chart Type | Design Tip |
|---|---|---|
| How are sales trending over the last 12 months? | **Line Chart** | Keep the line thick, remove gridlines. Do not use for non-time categories. |
| Which region had the highest sales? | **Bar Chart (Horizontal)** | Sort the PivotTable largest to smallest so the chart renders as a neat sorted staircase. |
| What is the breakdown of our 4 product categories? | **Donut or Pie Chart** | Never use pie charts for more than 5 categories. Add data labels showing percentages. |
| Target vs Actual Revenue by Store? | **Combo Chart** | Actuals as Columns. Targets as a Line with markers (no connecting line). |

**Chart Crimes to Avoid**

- **3D Charts:** Never use 3D pie or bar charts. They distort the data (the slice at the front looks artificially larger).
- **Clutter:** Delete the legend if you only have one metric (e.g., "Total Sales"). Use a clear chart title instead.
- **Axes starting above 0:** Bar charts must always have a Y-axis starting at 0, otherwise you exaggerate small differences.

# Part 13: Power Query (Automating Data Cleaning)

**What it is:** Power Query (PQ) is the ultimate ETL tool (Extract, Transform, Load) built right into Excel (Data tab > Get Data). If you spend 2 hours every Monday cleaning the same system export, PQ automates that to 1 click.

## The Magic of "Get Data From Folder"

**Scenario:** You manage a regional team. Every month, 12 city managers email you their sales in a CSV file. You have to open all 12, copy them, and paste them into one master file.

> **Practice Workshop: Folder Consolidation**
>
> 1. Put all 12 CSV files into one Windows folder (e.g., "2023_Sales").
> 2. Open a blank Excel workbook. Go to **Data > Get Data > From File > From Folder**.
> 3. Browse to "2023_Sales" and click Open.
> 4. A window shows all 12 files. Click the **Combine & Transform Data** button at the bottom.
> 5. The Power Query Editor opens. You will see a combined master table!
> 6. *Clean the data:* Right-click any unwanted column (like the file extension) and select **Remove**. Click the filter arrow on the Date column and uncheck "Null" to remove blank rows.
> 7. Click **Close & Load** (top left).
> 8. Excel loads the master table.
> 9. **The Magic:** Next month, when the 13th file arrives, drop it in the folder. Open this Excel file, Right-Click the table > **Refresh**. The new file is imported, cleaned, and appended instantly.

## Power Query vs Macros (VBA)

In the past, you had to write VBA code to automate cleaning. Today, Power Query is vastly superior for data tasks because:

- It requires zero coding (everything is UI buttons).
- It doesn't trigger corporate anti-virus warnings (unlike Macro .xlsm files).
- It processes millions of rows without crashing Excel.

# Part 14: Macros and VBA Basics

While Power Query handles data cleaning, Macros (VBA) handle *Excel interface automation*. E.g., formatting a report, saving it as PDF, and emailing it to the boss with one button click.

## Enabling the Developer Tab

Go to File > Options > Customize Ribbon. On the right side, check the box for **Developer**. Click OK.

> **Practice Workshop: Recording Your First Macro**
>
> **Task:** You receive a raw report daily. You always need to bold the headers, highlight them yellow, turn on filters, and freeze the top row.
>
> 1. Go to Developer Tab > **Record Macro**.
> 2. Name it: `FormatDailyReport` (No spaces allowed). Assign shortcut: `Ctrl + Shift + Q`. Click OK. *(Excel is now recording your clicks).*
> 3. Select Row 1. Make it Bold. Fill Color Yellow.
> 4. Go to Data > Filter (turns on filter arrows).
> 5. Go to View > Freeze Panes > Freeze Top Row.
> 6. Go to Developer Tab > **Stop Recording**.
>
> Tomorrow, open the new raw file, press `Ctrl + Shift + Q`, and the entire formatting applies instantly.

## Saving Macro Files

> **Critical Warning: File Extensions**
>
> If you record a macro and save the file as a standard `.xlsx`, **your macro will be permanently deleted** without recovery. You MUST go to File > Save As, and change the file type to **Excel Macro-Enabled Workbook (*.xlsm)**.

## Useful Ready-to-Use VBA Snippets

Press `Alt + F11` to open the VBA Editor. Insert > Module. Paste this code to unhide all sheets instantly.

```
Sub UnhideAllSheets()
    Dim ws As Worksheet
    For Each ws In ActiveWorkbook.Worksheets
        ws.Visible = xlSheetVisible
    Next ws
End Sub
```

# Part 15: Capstone Project - Interactive Sales Dashboard

**Goal:** Build a 1-page interactive dashboard combining everything you've learned. This is the ultimate test for an MIS Executive.

## Step 1: The Architecture (3 Tabs)

- **Tab 1 (Raw Data):** Paste your transaction data here. Format as Ctrl+T Table. Hide this tab later.
- **Tab 2 (Calculations):** Insert 4 PivotTables here based on Tab 1. Space them 10 rows apart.
  - Pivot 1: Total Sales & Profit by Month.
  - Pivot 2: Sales by Region.
  - Pivot 3: Top 5 Sales Reps.
  - Pivot 4: Sales by Product Category.
- **Tab 3 (Dashboard):** Go to View > Uncheck Gridlines. This is your blank canvas.

## Step 2: Building the Visuals

1. Go to Tab 2 (Calculations). Click Pivot 1. Insert > Line Chart. Cut (Ctrl+X) the chart, go to Tab 3 (Dashboard), and Paste (Ctrl+V).
2. Repeat for Pivot 2 (Donut Chart), Pivot 3 (Horizontal Bar Chart), and Pivot 4 (Column Chart).
3. Arrange them cleanly on the Dashboard tab. Ensure they align perfectly.

**Step 3: Connecting the Interactivity (Slicers)**

1. Click any chart on your Dashboard tab.
2. Go to PivotChart Analyze > **Insert Slicer**. Check the boxes for "Year" and "Region". Click OK.
3. Two interactive button panels appear. *But they only control one chart right now.*
4. Right-click the "Region" Slicer panel > **Report Connections**.
5. Check the boxes for all 4 PivotTables in your file. Click OK. Repeat for the "Year" Slicer.
6. **Result:** When your manager clicks "West" on the Slicer, every single chart on the dashboard instantly updates to show only West region data.

## Part 16: Job Interview Preparation & Assessment

Most corporate interviews for Analyst or MIS roles require passing a timed, 30-minute practical laptop test. Below is a simulation of a standard assessment.

### Mock Practical Test (30 Minutes)

**The Data Provided:** A raw dump of 10,000 rows. Columns: `OrderDate, EmpID, ClientName_City (Combined), Category, SalesAmt`.

1. **Task 1: Clean the Client Column (5 mins)**
   *Prompt:* The client name and city are stuck together (e.g., "Infosys_Bengaluru"). Split them into two columns.
   *Solution:* Insert a column next to it. Select the data, go to Data > Text to Columns. Choose 'Delimited', check 'Other' and type the underscore `_`. Finish. Rename headers.
2. **Task 2: Lookup Missing Data (5 mins)**
   *Prompt:* You are provided a second sheet with EmpID and Employee Names. Pull the Names into your main table.
   *Solution:* Create a new column. Use `=VLOOKUP(B2, Sheet2!$A$1:$B$50, 2, 0)`. Double-click the fill handle to apply to all 10,000 rows.
3. **Task 3: Conditional Formatting (5 mins)**
   *Prompt:* Highlight any SalesAmt greater than ₹1,00,000 in Green.
   *Solution:* Select SalesAmt column. Home > Conditional Formatting > Highlight Cells Rules > Greater Than... Type 100000, select Green Fill.
4. **Task 4: Reporting via Pivot (10 mins)**
   *Prompt:* Which Employee generated the highest total sales in Bengaluru?
   *Solution:* Convert data to Table. Insert PivotTable. Drag City to Filters (Filter to Bengaluru). Drag Emp Name to Rows. Drag SalesAmt to Values. Sort Values Largest to Smallest. Give the recruiter the top name.

**Top 3 Reasons Candidates Fail the Test**

1. **Not locking references:** Using VLOOKUP without `$`, causing #N/A errors halfway down the sheet.
2. **Ignoring Formatting:** Providing a PivotTable with raw numbers like 1450394 instead of formatted currency ₹14,50,394. Presentation matters.
3. **Panic on #DIV/0! Errors:** If calculating a percentage year-over-year and previous year is 0, it throws an error. Wrap it in `=IFERROR(formula, 0)` to look professional.

## End of Intensive Practice Edition

Master these principles, and you are ready for any corporate Excel challenge.